# Neutral-current Higgs-related $4b$ production

FCC-ue, unpolarized 1 TeV benchmark | Preliminary LO results

**Beam configuration:** $E_{\mu^+} = 2000$ GeV, $E_{e^-} = 120$ GeV; $\sqrt{s} = 979.796$ GeV; $P_{\mu^+} = P_{e^-} = 0$.
**Generator:** MadGraph5_aMC@NLO 3.6.4, LO, Standard Model.

## Processes and cross sections

| Partonic production channel | Before $b$ cuts [fb] | After $b$ cuts [fb] |
|---|---|---|
| $\mu^+e^- \to \mu^+e^- HH \to \mu^+e^- 4b$ | 0.00643724 | $0.00638240 \pm 0.00000990$ |
| $\mu^+e^- \to \mu^+e^- b\bar{b}H \to \mu^+e^- 4b$ | 0.02040127 | $0.01292500 \pm 0.00005939$ |

## Fiducial definition

- **Offline $4b$ selection:** four massive bottom partons with $|\eta_b| < 5$; there must exist two disjoint opposite-charge $b\bar{b}$ pairings with $110 \leq m_{b\bar{b}} \leq 130$ GeV for both pairs. No minimum $b$-parton $p_T$ or $\Delta R$ cut.

**Audit trace:** 4 diagrams ($HH$); 86 diagrams ($b\bar{b}H$). Both MadGraph runs and offline analyses completed; results remain preliminary. Distinct outgoing charged-lepton states do not interfere with the previously studied charged-current neutrino final state.